

Improving Preparedness of Communities for Evacuations using ZEVs

Planning Infrastructure Investments

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December 05, 2024

Introduction

Smart Power Lab

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Field of Research:

- **Power Systems Planning, Operation, and Control**
- **Power Electronics Converters**
- **Microgrids Control and Stability**
- **Renewable Energy Sources and Electric Vehicles**

Outline

Overview of the Project Procedure

Operation and Resilience

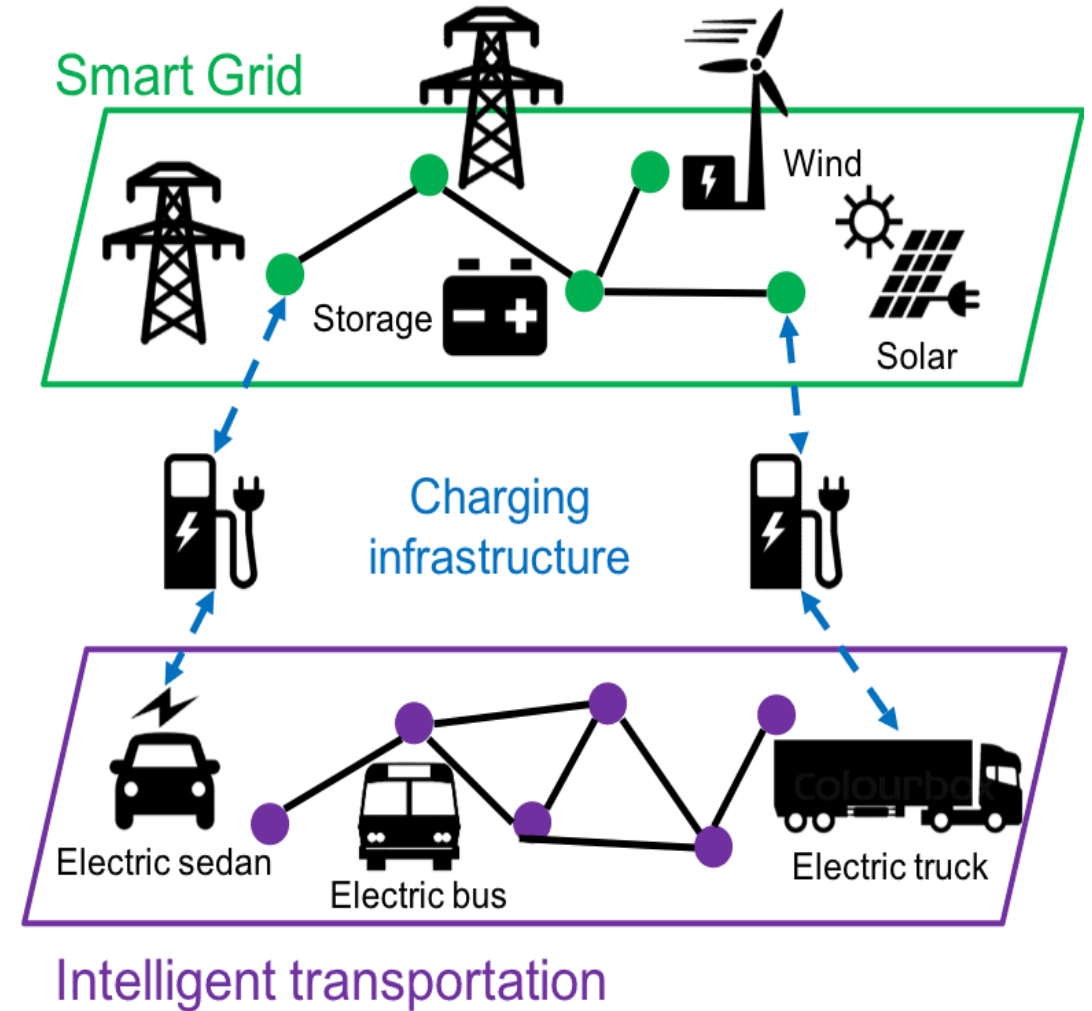
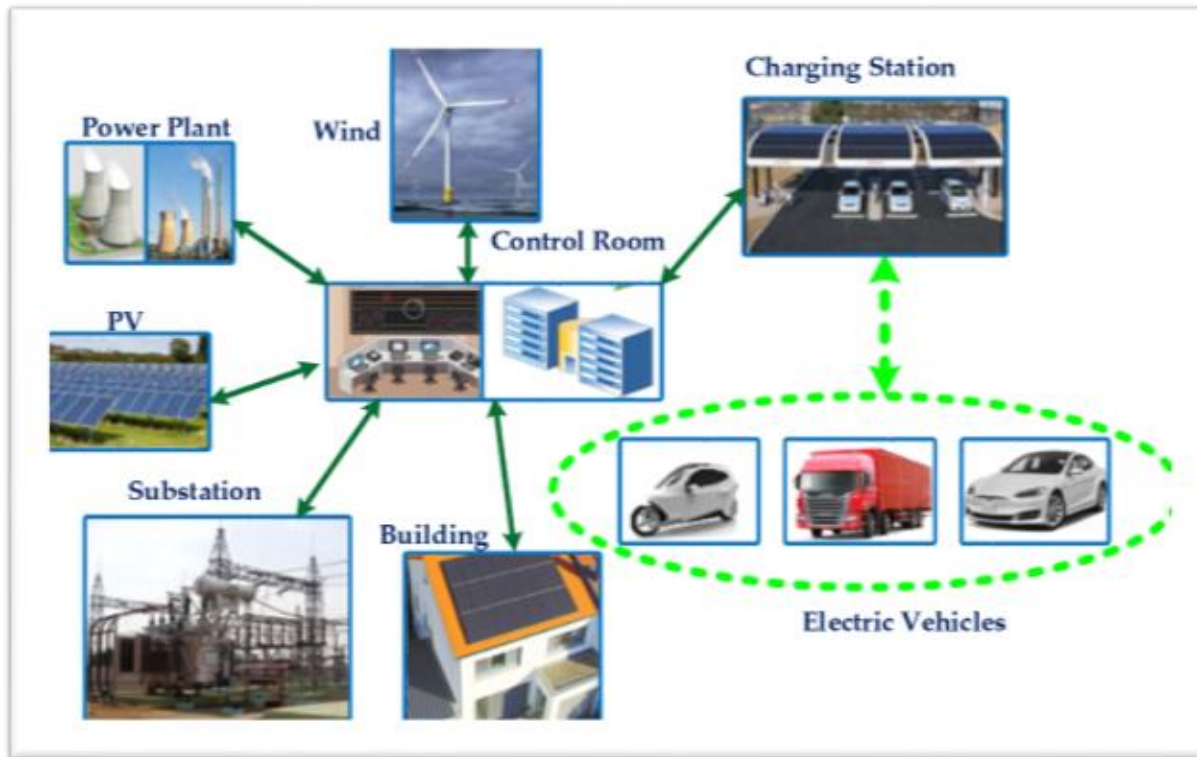
Infrastructure Investment

Evacuation Scenario

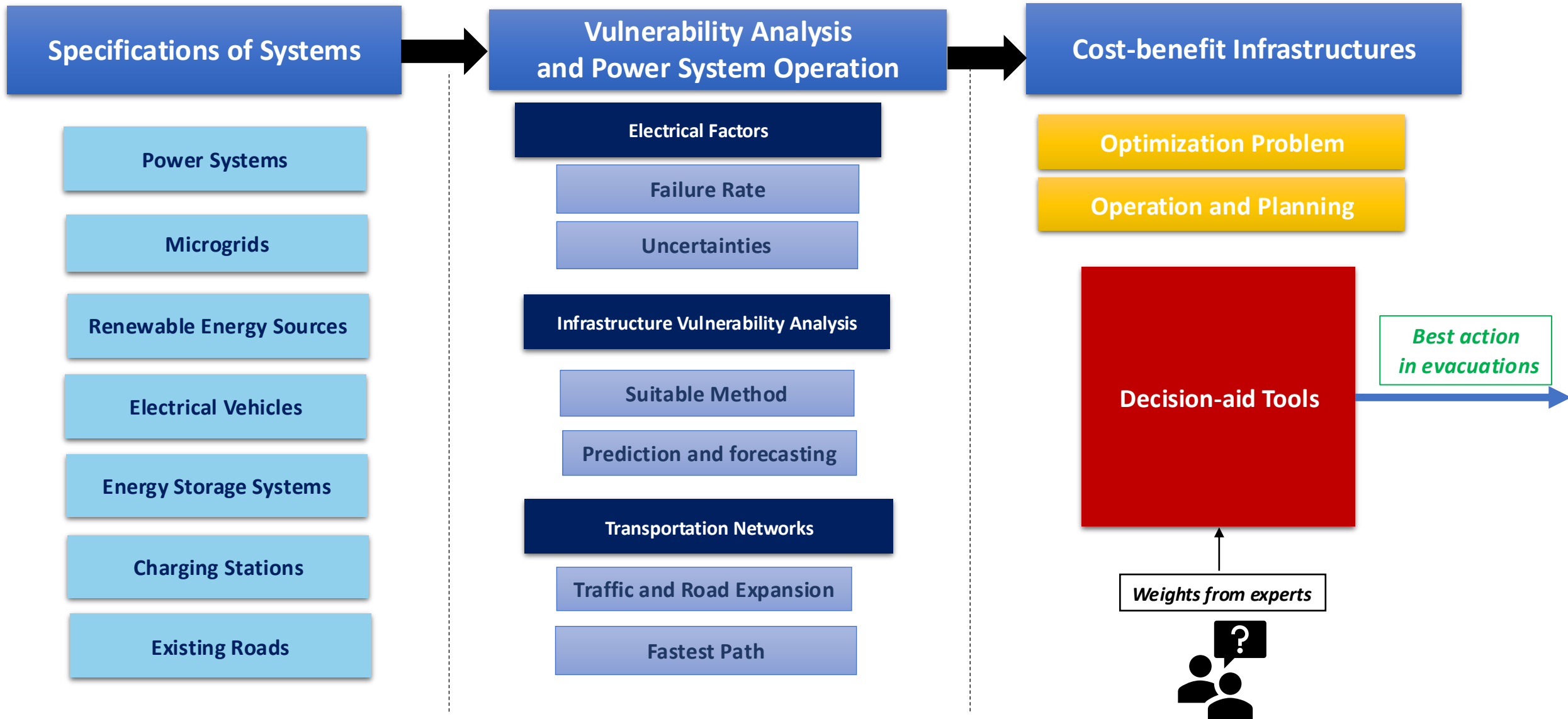
Results and Discussion

Fact Sheet

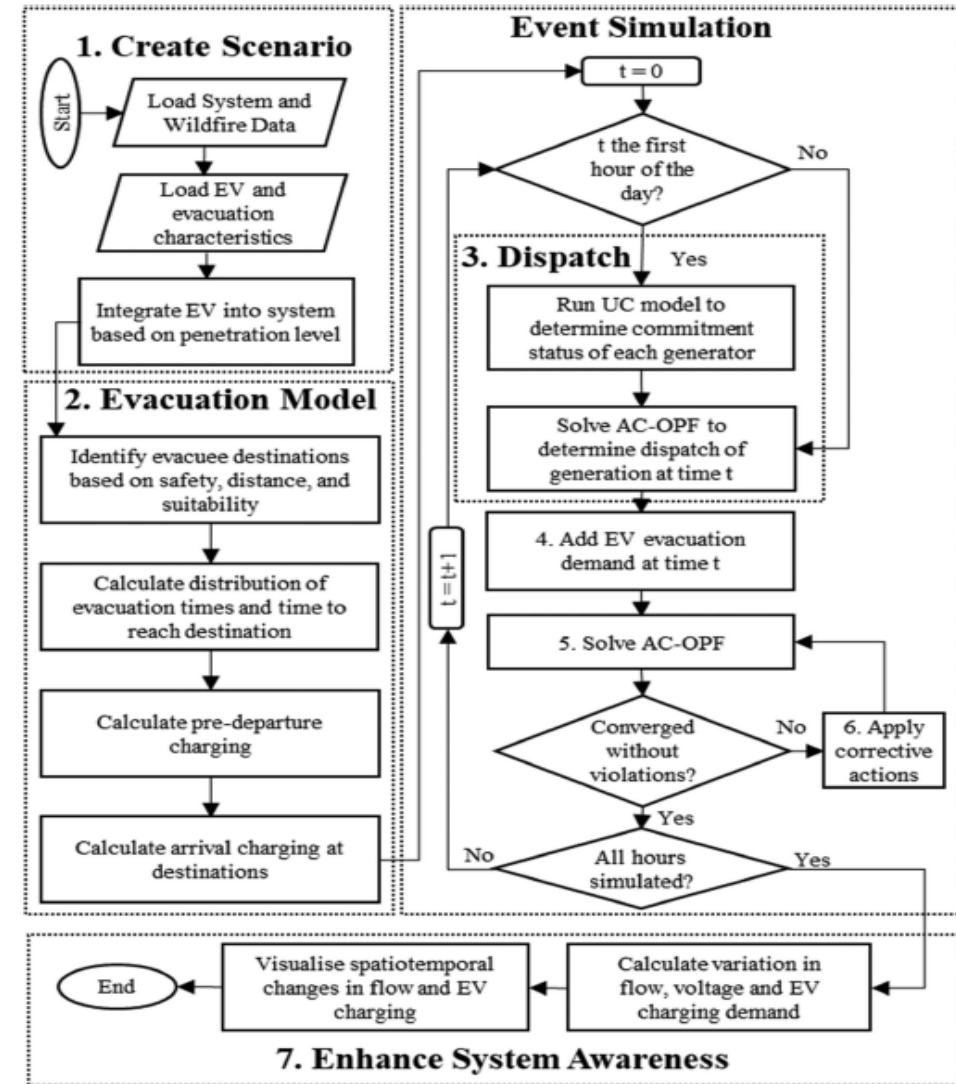
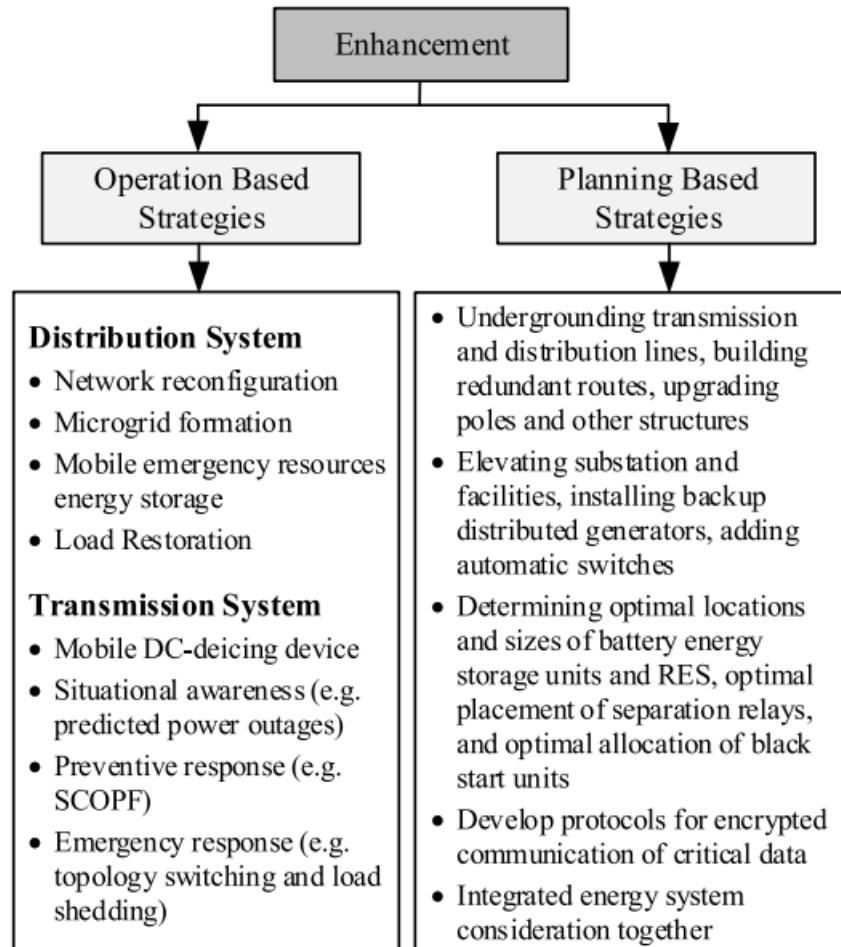
Structure of the Systems: Power and Transportation



Overview of Task #2



Power Systems Studies



Case Study

Use cases: 7-bus, 24-bus, and 37-bus power system

Buses:

Number of Buses: 24

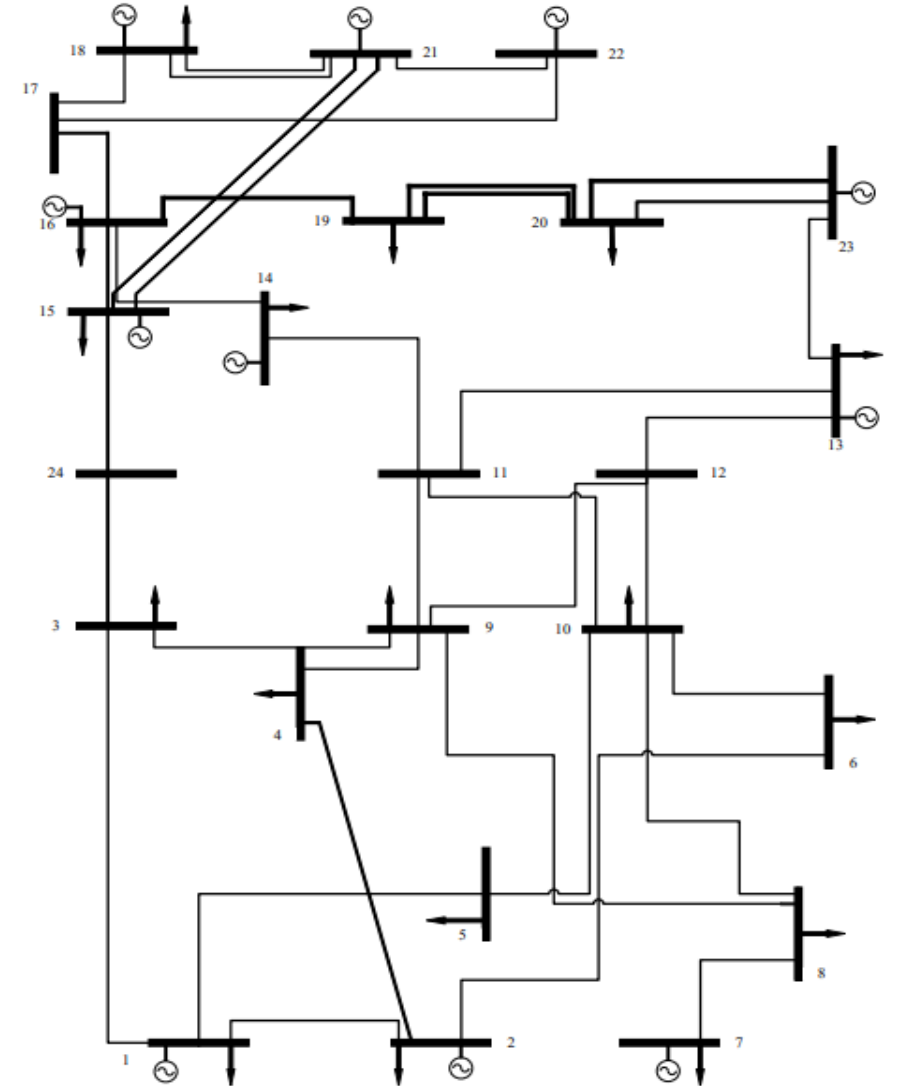
Bus Types: Includes slack (reference), generator, and load buses

Voltage Levels:

Multiple voltage levels: Common ones might be 138 kV and 230 kV

Generators:

Locations: Specified at certain buses (not all buses have generators)



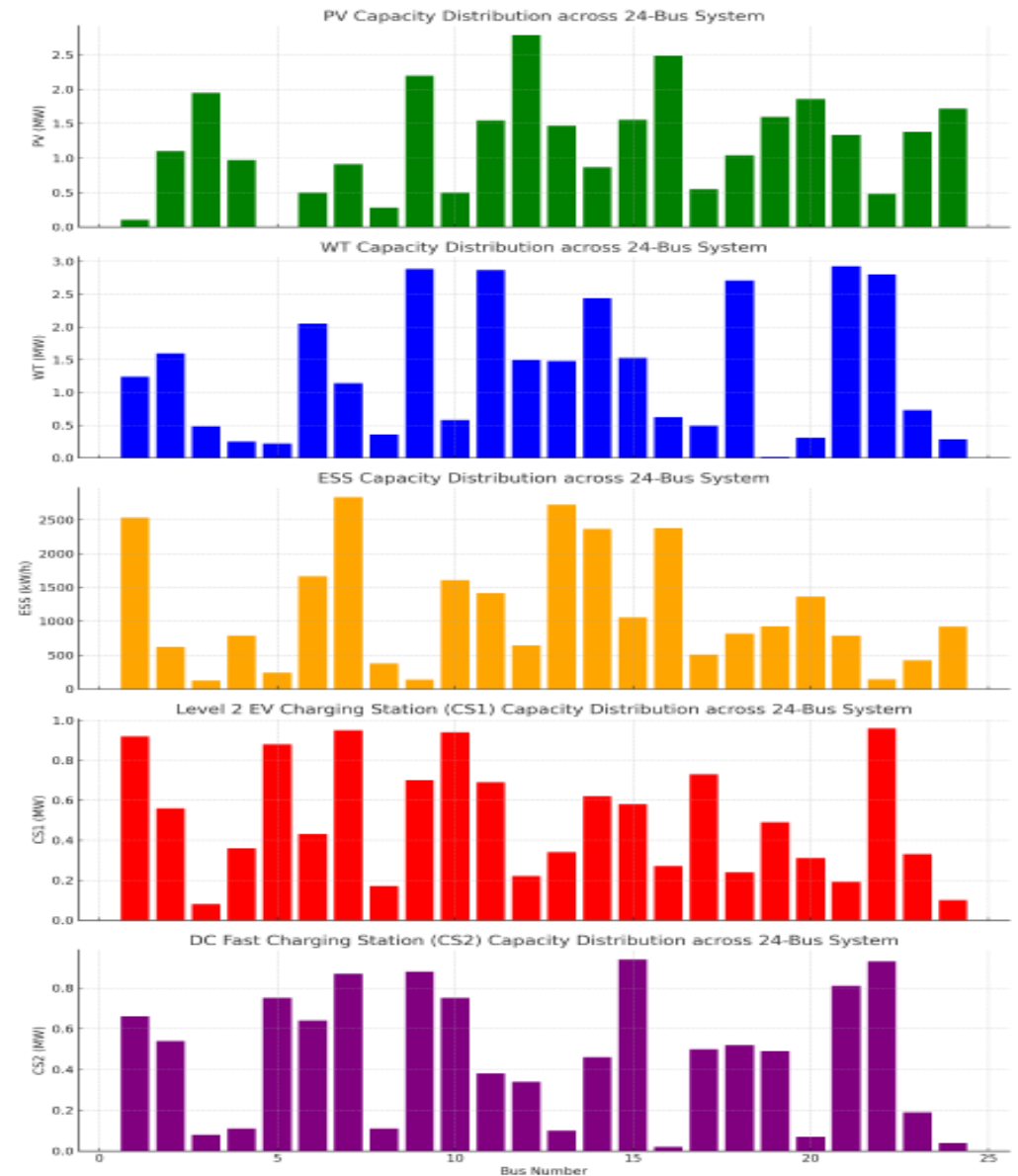
Infrastructure Investment

DESIRABLE INFRASTRUCTURES AND THEIR SPECIFICATIONS

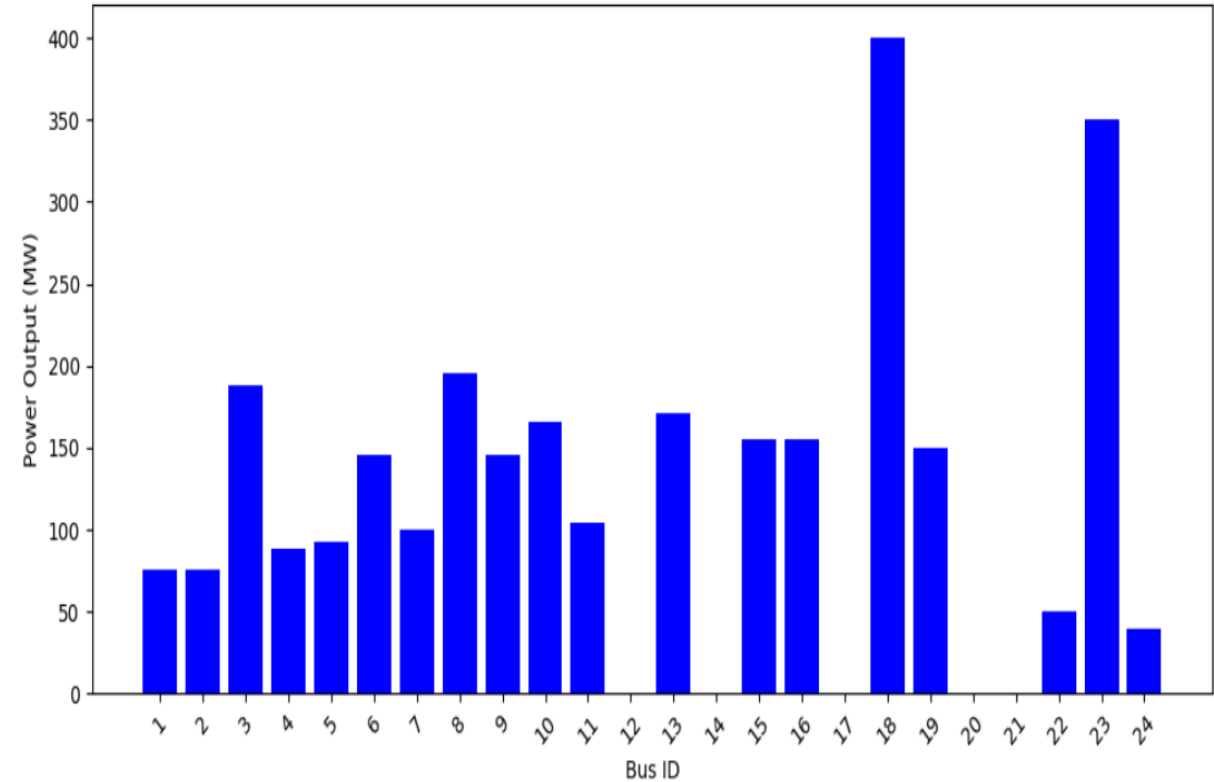
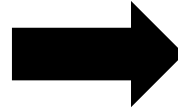
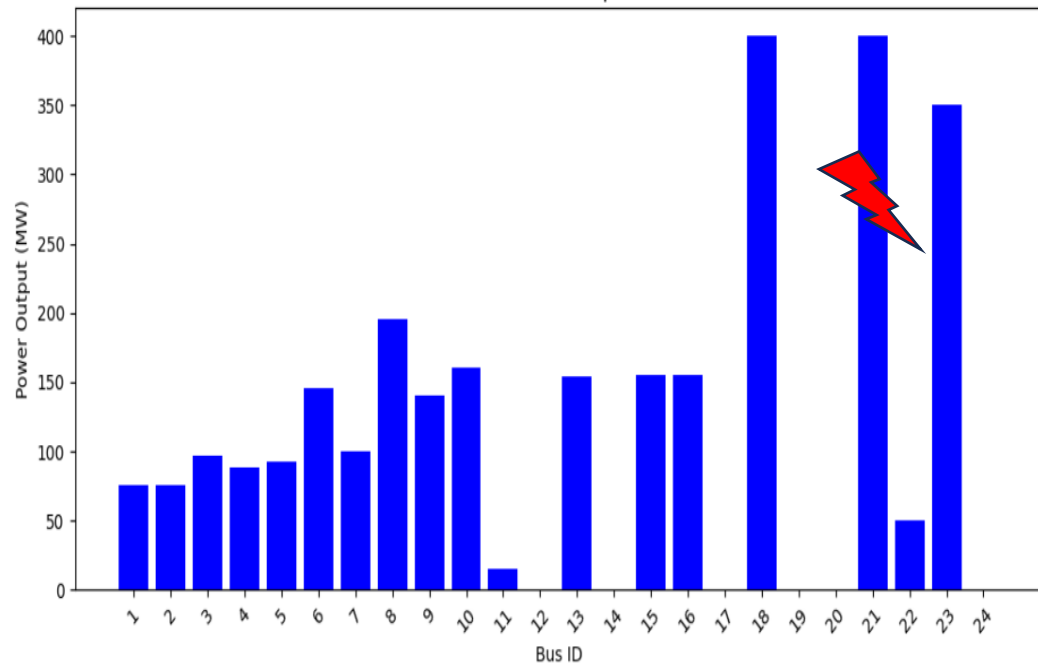
Infrastructure	Investment Cost (k\$/MW)	Operation Cost (k\$/MW/year)	Capacity (MW)	Efficiency (%)	Lifetime (years)
PV System	1000	10	2	18	25
WT	1500	20	2	35	20
ESS	500	15	5	90	15
Level 2 EV Charging Station (CS ₁)	500	5	0.022	95	10
DC Fast Charging Station (CS ₂)	1000	10	0.35	95	10

DEGRADATION RATE AND CO₂ EMISSIONS REDUCTION OF INFRASTRUCTURES

Infrastructure	Degradation Rate (%/year)	CO ₂ Emissions Reduction (tons/MW/year)
PV System	0.5	80
WT	0.3	60
ESS	2	0
Level 2 EV Charging Station (CS ₁)	1	5
DC Fast Charging Station (CS ₂)	1	10



Results and Discussion (Optimal Power Flow)



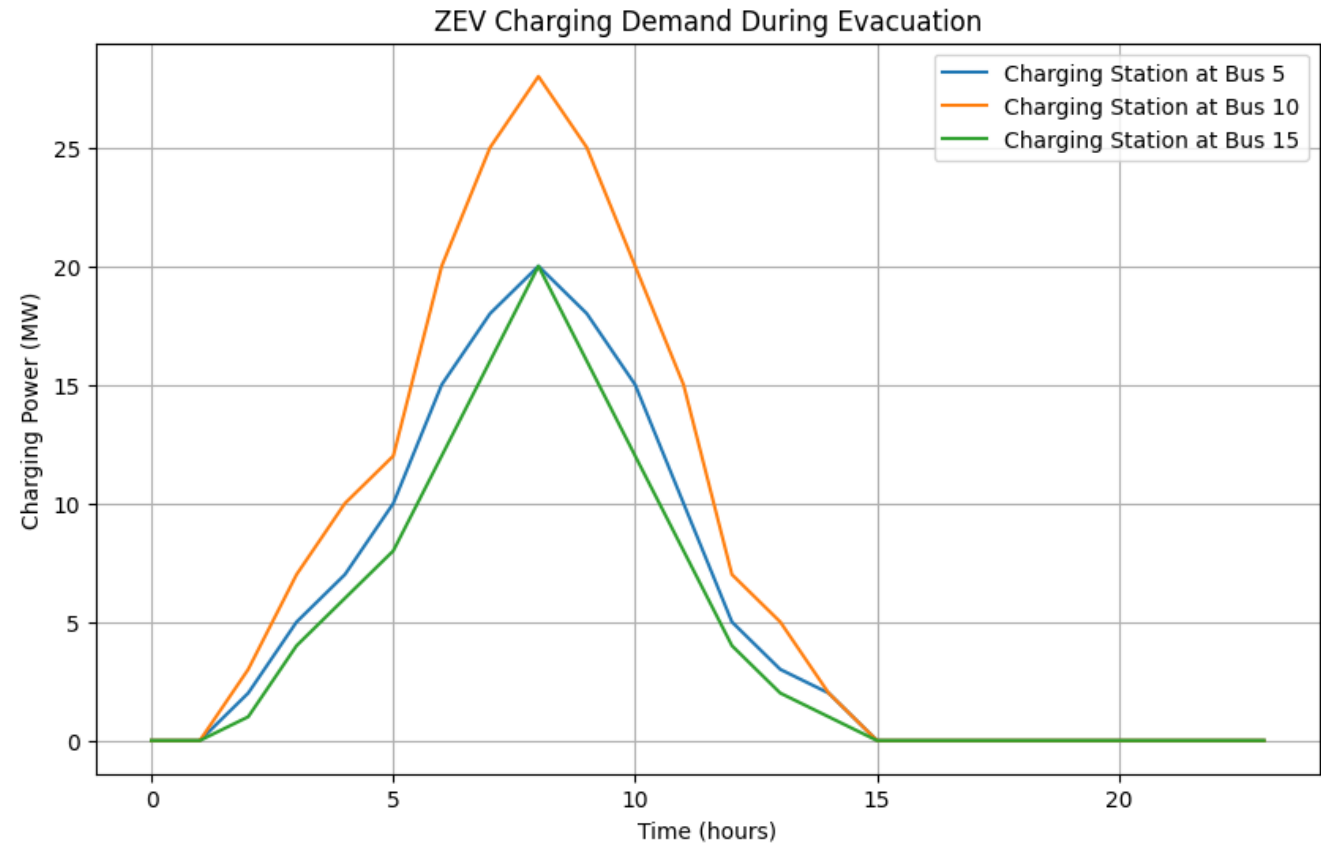
N-1: Generator #21 Fails

Evacuation Scenario - 1

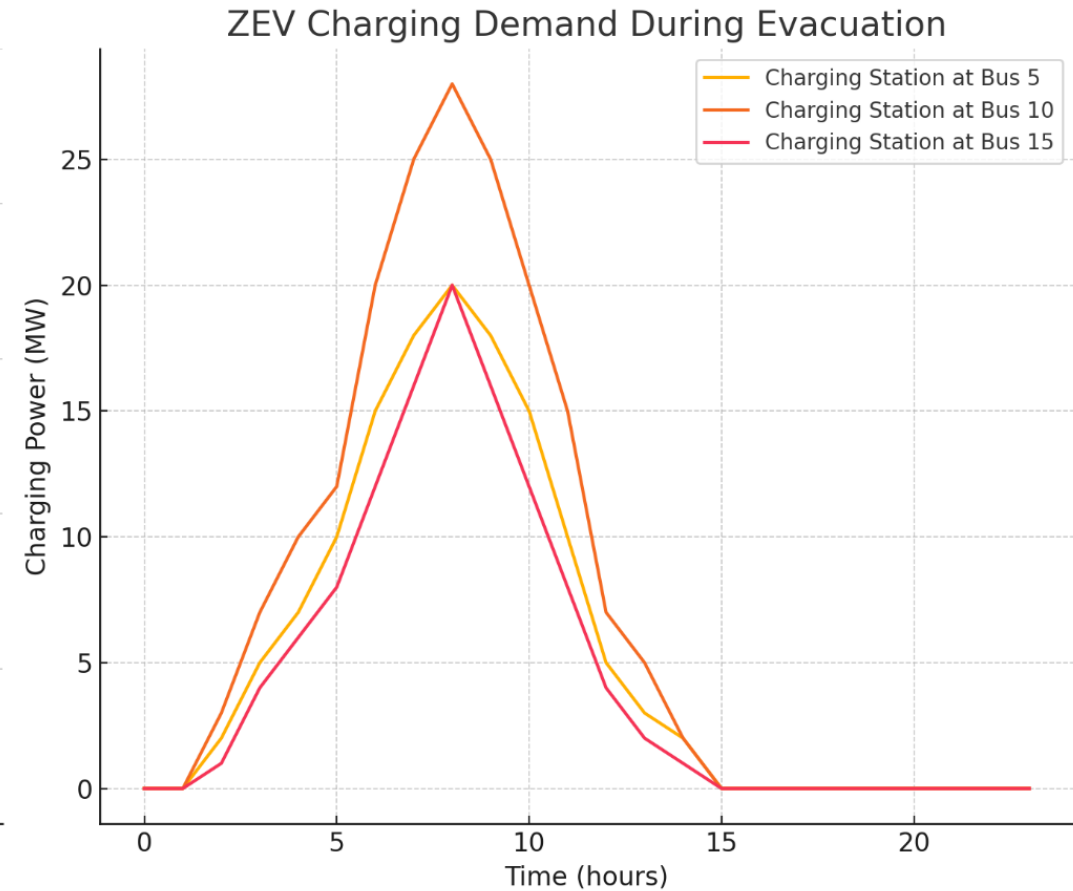
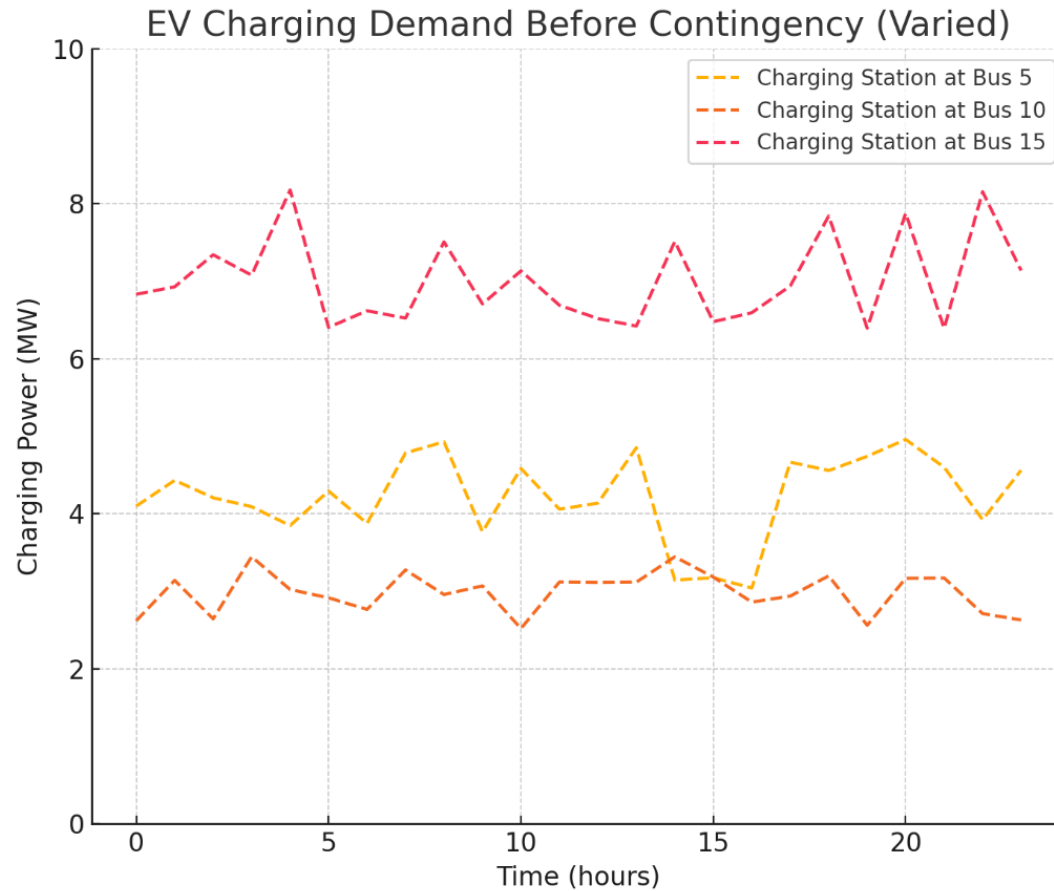
Scenario 1: Minimal ZEV Adoption Limited grid stress, underutilized infrastructure. Highlighted need for adaptive load management.

Scenario 2: Moderate ZEV Penetration Increased grid complexity and charging demand. Real-time OPF managed demand but revealed bottlenecks.

Scenario 3: High ZEV Adoption Significant grid challenges: voltage drops, power losses. Renewable integration and ESS improved resilience.



Evacuation Scenario - 2



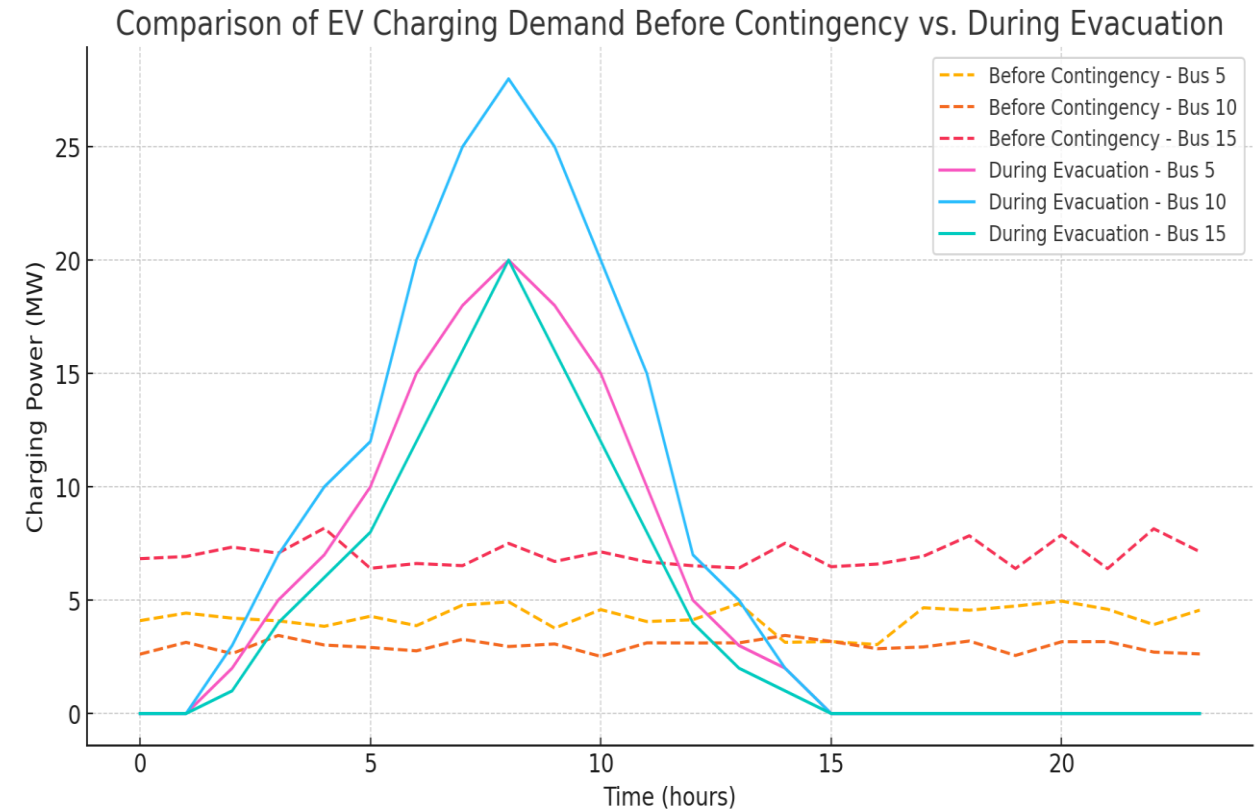
Evacuation Scenario - 3

Evacuation Impact: Sharp demand peaks highlight stress on the grid during evacuation. Concentrated charging period requires advanced load management.

Grid Challenges: High demand can lead to instability without proper management. Real-time OPF and demand response strategies are essential.

Recommendations:

- ✓ Increase charging station capacity at high-demand buses (Infrastructure investment).
- ✓ Deploy Energy Storage Systems (ESS) for peak demand buffering.
- ✓ Use AI tools for demand forecasting and real-time adjustments.



Community Partners

Challenges in the Project:

1. Data Gaps:

- Lack of detailed real-time ZEV charging demand and grid data.
- Limited access to historical outage and emergency response records.

2. Uncertainty in Evacuation Scenarios:

- Highly dynamic ZEV loads and unpredictable demand spikes.
- Need for robust models to simulate real-world evacuation scenarios.

3. Integration Issues:

- Balancing ZEV charging with grid stability and renewable integration.

Community Partners

Data Requests:

1. ZEV Charging Infrastructure and Energy Need:

- Locations and capacities of existing charging stations.
- Charging demand patterns during normal and emergency conditions.

2. Power System Resilience:

- Historical data on power outages during emergencies.
- Current capacities and response times of energy storage systems.
- availability of renewable energy sources at different locations.

3. Evacuation Scenarios: Records of previous disaster and evacuation events involving ZEVs.

4. Emergency Management Criteria: Key factors for prioritizing infrastructure investments and constraints during evacuations.

Q&A?

Thanks for your attention!

My LinkedIn:



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